Five reactant complexes are necessary and sufficient for absolute concentration robustness with bistability

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Abstract. The exact minimum number of distinct reactant complexes for coexistence of absolute concentration robustness (ACR) and two exponentially stable positive equilibria in an at-most-bimolecular mass-action network is five. The lower bound combines the earlier dimension-two obstruction with a complete four-reactant exclusion: an exact finite classification reduces the latter to nineteen scalar normal forms, each having at most two nondegenerate equilibria with opposite determinant signs. The upper bound is an explicit four-species, six-reaction network with integer rates and three rational equilibria in one compatibility class, two of which are stable. All computational claims have exact reproducible certificates. The argument and the certificates were rechecked by a fresh-context verifier; there has been no human peer review.

Question, result and scope

Kaihnsa, Nguyen and Shiu [1], Section 6, write:

> *"Another interesting direction, therefore, is to investigate conditions for coexistence of ACR and multistability, rather than multistationarity."*

They conjecture stricter size bounds for this coexistence. The earlier repository manuscript [2] establishes dimension at least three, and bounds the number of reactant complexes between four and six. Its remaining four-reactant cases, called S3, S7 and S9, were excluded only numerically. Our precise follow-up question is the exact minimum number of reactant complexes. We determine it.

Here every source and every product has total molecularity at most two. The zero complex is allowed and counts as a complex, also as a reactant complex: Example 4.13 of [1] counts $0 \to X$ among its four reactant complexes. Rate constants are strictly positive. ACR means that at least one positive equilibrium exists and a fixed species has the same concentration at *all* positive equilibria, including equilibria in different classes. Stability is exponential stability relative to the stoichiometric compatibility class. Arbitrarily many reactions from each source are allowed; we do not assume one reaction per source.

Theorem 1. Among mass-action systems with at-most-bimolecular reactant and product complexes, the minimum number of distinct reactant complexes allowing ACR and two exponentially stable positive equilibria in one stoichiometric compatibility class is exactly five.

The lower and upper bounds concern arbitrary numbers of species and reactions. The five-source construction uses four species; no claim of simultaneous minimization of all

network sizes is made.

Theorem 2. Let a mass-action system have stoichiometric rank three, four distinct reactant complexes, and at-most-bimolecular sources and products. If it has ACR, each positive compatibility class contains at most two nondegenerate positive equilibria. When there are two, their restricted Jacobian determinants have opposite signs. In particular, at most one is exponentially stable.

Corollary 3. An at-most-bimolecular mass-action system with ACR and two exponentially stable positive equilibria in one class has at least five distinct reactant complexes.

Proof. With r reactant complexes a positive equilibrium gives a strictly positive vector in the kernel of the aggregated coefficient matrix N, so $\operatorname{rank} N \le r-1$. At a nondegenerate equilibrium, $\operatorname{im} N$ equals the stoichiometric subspace S: indeed $\operatorname{im} J \subseteq \operatorname{im} N \subseteq S$, while $J|_S$ is invertible. For $r \le 4$ this forces $\dim S \le 3$. Dimension two is excluded by the dimension-two theorem of [2], which holds for every number of species and reactant complexes. Dimension one is excluded by Proposition 3.10 of [1]; directly, on a class the field is a fixed vector times a polynomial of degree at most two in the class coordinate, and two simple zeros of such a polynomial have opposite derivative signs. Dimension zero has singleton classes. Dimension three forces $r = 4$ and is excluded by Theorem 2. □

The rank-three theorem is proved here without using the earlier dimension-two theorem. Only the corollary uses that earlier result. The following construction attains the bound and, together with the corollary, proves Theorem 1.

1 A five-source network attaining the minimum

Consider four species A, B, C, R and the six reactions

$$0 \xrightarrow{64} C, \qquad A+B \xrightarrow{3} 2B, \qquad A+C \xrightarrow{5} B+C, \qquad B+C \xrightarrow{64} A, \\ 0 \xrightarrow{1} R, \qquad R \xrightarrow{1} 0. \tag{4}$$

Its five distinct reactant complexes are $0, A+B, A+C, B+C, R$. Every reactant and product has molecularity at most two. With the mass-action convention kx^y its equations are

$$\begin{aligned} \dot a &= -3ab - 5ac + 64bc, & \dot b &= 3ab + 5ac - 64bc, \\ \dot c &= 64 - 64bc, & \dot r &= 1 - r. \end{aligned} \tag{5}$$

The stoichiometric subspace is $\{v : v_a + v_b = 0\}$, of dimension three; the only independent conservation law is $a + b = T$. Every positive equilibrium has $r = 1$, so ACR holds globally, across all compatibility classes. In fact arbitrary positive rates in the last two reactions give the ACR value k_{0R}/k_{R0}.

In the class $a + b = 9$, equilibrium requires $b = 9 - a$, $c = 1/(9-a)$ and $r = 1$, with $0 < a < 9$. The remaining equation factors exactly as

$$\dot a\big|_{b=9-a,\, c=1/(9-a)} = \frac{-3(a-4)(a-6)(a-8)}{9-a} = 0.$$

Thus the complete list of positive equilibria in this class is

$$P_1 = (4, 5, 1/5, 1), \qquad P_2 = (6, 3, 1/3, 1), \qquad P_3 = (8, 1, 1, 1).$$

In the class coordinates (a, c, r) the restricted Jacobian is

$$J_{\mathrm{red}} = \begin{pmatrix} 6a - 27 - 69c & 576 - 69a & 0 \\ 64c & -64(9-a) & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Its characteristic polynomial is $(\lambda + 1)(\lambda^2 - \tau\lambda + \Delta)$, where τ and Δ are the trace and determinant of the upper left 2×2 block:

Equilibrium	τ	Δ	Stability in the class
P_1	$-1684/5$	1536	Exponentially stable
P_2	-206	-768	Saddle
P_3	-112	1536	Exponentially stable

The trace-negative, determinant-positive criterion for a real 2×2 matrix proves the two stability assertions exactly; no numerical eigenvalue calculation is used.

The regulator can also affect the other species without increasing the number of sources. Replace $0 \xrightarrow{64} C$, $0 \xrightarrow{1} R$, $R \xrightarrow{1} 0$ by $0 \xrightarrow{62} C$, $0 \xrightarrow{1} C + R$, $R \xrightarrow{1} C$. Only $\dot{c}$ changes, by adding $r - 1$. The equilibria are unchanged and the Jacobian remains triangular in the two blocks, with exactly the same characteristic polynomial. Both versions are verified directly from their reaction lists by `certify_five_example.py`.

2 Toric reduction and the determinant sign

List the sources as $y_1, \ldots, y_4$, and write $f(x) = Nm(x)$, where $m_a(x) = x^{y_a}$. It suffices to consider a system possessing a nondegenerate positive equilibrium x^*. Then $\operatorname{rank} N = 3$ and $\ker N$ is generated by $m(x^*) > 0$. Let A be the $3 \times n$ matrix with rows $y_a - y_4$, $a = 1, 2, 3$. The equilibrium equations are

$$A \log x = A \log x^*, \qquad \mathcal{E} = x^* \circ \exp(\ker A).$$

Nondegeneracy forces $\operatorname{rank} A = 3$. Otherwise the differential of these equations, and hence the restricted Jacobian, has rank at most two. ACR in species i is equivalent to $e_i \in \operatorname{row}(A)$, or to a zero ith row in any basis matrix V of $\ker A$.

Let W have rows forming a basis of $S^{\perp}$. Compatibility is $Wx = Wx^*$. Set $N_0 = [N_1\ N_2\ N_3]$ and $p_a = m_a(x^*)/m_4(x^*) > 0$. Any three columns of N are independent. At equilibrium,

$$J = m_4(x) N_0 \operatorname{diag}(p_1, p_2, p_3) A \operatorname{diag}(1/x).$$

Consequently, for a fixed nonzero constant σ and a positive factor $q(x)$,

$$\det(J|_S) = \sigma q(x) \det \begin{pmatrix} W \\ A \operatorname{diag}(1/x) \end{pmatrix}. \tag{6}$$

To see this, use the basis N_0 for S and complete it by columns B to a basis of $\mathbb{R}^n$. Multiplication of the displayed block matrix by $[B\ N_0]$ gives a block triangular matrix with diagonal blocks WB and $A \operatorname{diag}(1/x) N_0$. All change-of-basis determinants are fixed and nonzero.

The same determinant can be written with a basis matrix V of $\ker A$. Choose C with AC invertible. Then $[V\ C]$ is invertible, and multiplying the block matrix on the right by $\operatorname{diag}(x)[V\ C]$ gives a block triangular matrix with diagonal blocks $W\operatorname{diag}(x)V$ and AC. Hence, for a fixed constant $c \neq 0$,

$$\det\begin{pmatrix} W \\ A\operatorname{diag}(1/x) \end{pmatrix} = \frac{c}{x_1 \cdots x_n} \det(W\operatorname{diag}(x)V). \tag{7}$$

At a positive equilibrium, nondegeneracy is therefore equivalent to $\det(W\operatorname{diag}(x)V) \neq 0$, and the sign of $\det(J|_S)$ is a fixed sign times the sign of $\det(W\operatorname{diag}(x)V)$.

Replacing logarithmic binomials by ordinary binomials multiplies their differential rows by positive monomial factors, followed by a fixed nonsingular row transformation. Fixing the ACR concentration and performing linear eliminations therefore preserves the comparison of determinant signs. In a one-variable reduction $F(t) = 0$, the augmented determinant is a fixed sign times a positive factor times $F'(t)$, provided the eliminated equations have nonsingular differentials on the positive parameter interval. We verify this condition explicitly below. Two simple roots of a quadratic on an interval have opposite derivative signs. The same holds for $h(t) - p$ when $p > 0$ and $\log h$ is strictly concave on that interval. Since a real Hurwitz 3×3 matrix has negative determinant, opposite determinant signs preclude two stable equilibria.

3 Exhaustive source classification

A species absent from every source has a nonnegative derivative. At a positive equilibrium every reaction rate is positive, so no reaction can produce that species either. Such species are inert and can be removed. They cannot be ACR species, since their arbitrary positive concentrations give equilibria with different values. Four bimolecular sources then involve at most eight species.

The source classification is elementary and finite. For $4 \leq n \leq 8$, enumerate four-element subsets of

$$\{0, e_i, 2e_i, e_i + e_j : 1 \leq i < j \leq n\}.$$

Retain only sets covering all species, having rank $A = 3$, and having an ACR coordinate. The rank test uses integer 3×3 minors. The ACR coordinates are precisely the indices contained in every nonzero three-column minor: deleting such a column lowers the rank to two. Canonicalization permutes the four sources and sorts the species columns, so no floating-point rank decision or heuristic is involved. The script `certify_sources.py` performs this enumeration.

For three species the toric equilibrium set is a singleton. For four species there are 26 source types. In each, a generator v of $\ker A$ has at most three distinct entries, including zero; this is checked by its integer cofactor formula. The class equation is

$$\sum_{j=1}^{4} W_j x_j^* e^{v_j t} - W x^* = 0.$$

It has at most three distinct exponential terms. A nonzero sum of three real exponentials has at most two real zeros counted with multiplicity: divide by the smallest exponential and apply Rolle's theorem to the resulting derivative, which has at most two terms. If the

sum is identically zero, all equilibria on the curve are degenerate. In the nondegenerate case the two determinants have opposite signs by (6).

For five or more species the exact list, up to species and source permutations, is as follows.

Type	Sources	ACR species
S1	$0, E, C+D, A+B$	E
S2	$0, 2E, C+D, A+B$	E
S3	$E, 2E, C+D, A+B$	E
S4	$E, D, C+E, A+B$	C
S5	$E, D+E, C+E, A+B$	C, D
S6	$E, D+E, C+D, A+B$	D
S7	$E, D+E, 2C, A+B$	D
S8	$E, D+E, B+C, A+C$	D
S9	$F, E+F, C+D, A+B$	E

There are eight types on five species, one on six species, and none on seven or eight. This reproduces, in fully exact arithmetic, the earlier source list in [2].

For S4, S5, S6 and S8 the varying equilibrium concentrations, after positive constant factors are removed, belong to three groups $g_1 = u, g_2 = w/u, g_3 = w$. For S4, c is fixed and d, e are proportional to ab; for S5, c, d are fixed and e is proportional to ab; for S6, d is fixed and c, e are proportional to ab; for S8, d is fixed, a, b are proportional, and e is proportional to ac. The class equations are $Qg = T$, two linear equations in the three group variables. If $\operatorname{rank} Q < 2$, the equilibria are degenerate. Otherwise write the affine solution line as $g(t) = g^0 + t\rho$, where ρ spans $\ker Q$. Positivity of all three components cuts out an interval. The remaining equation is the polynomial $F(t) = g_1(t)g_2(t) - g_3(t)$ of degree at most two. The augmented determinant with rows Q and $\nabla(g_1 g_2 - g_3)$ is a fixed nonzero multiple of $F'(t)$: the cross product of the rows of Q is a fixed nonzero multiple of ρ. The group embedding is linear with independent columns and positive constant entries on disjoint coordinate groups. Eliminating the other fixed linear equations therefore changes the original augmented determinant only by a fixed nonzero factor. Hence (6) applies, and two nondegenerate equilibria have opposite determinant signs. If F vanishes identically, every equilibrium is degenerate.

4 Exhaustive stoichiometric classification

We give a fresh exact classification for S1, S2, S3, S7 and S9. It supersedes the numerical subspace search for the three previously unresolved cases.

For a fixed source set, form every possible nonzero reaction vector $p - y_a$, where p is any at-most-bimolecular product. Every rank-three stoichiometric subspace is spanned by three of these vectors. Enumerate these triples and identify equal spans by their primitive exterior coordinates (their 3×3 minors, divided by their gcd, with the first nonzero entry positive). For each span S, retain all allowed reaction vectors lying in it, giving a maximal matrix G. Membership is checked by the integer adjugate identity in a nonsingular three-coordinate projection.

Lemma 8. If some subnetwork with this span has a positive equilibrium, the maximal matrix G has a strictly positive kernel vector. In a three-coordinate projection, failure of this condition is detected by a nonzero cross product of two columns having nonnegative inner products with every column, after possibly reversing its sign.

Proof. The subnetwork equilibrium supplies a positive dependence among vectors spanning S. Their positive cone is therefore all of S: any linear representation can be made nonnegative by adding a sufficiently large multiple of that dependence. Each additional vector can thus be cancelled by a nonnegative combination of the old vectors. Adding these dependencies to the original positive dependence gives a dependence positive on every maximal-network reaction.

Conversely, a finite collection spanning $\mathbb{R}^3$ positively spans $\mathbb{R}^3$ exactly when its dual cone is $\{0\}$. If this dual cone is nonzero it is pointed, since the original vectors span $\mathbb{R}^3$. It has an extreme ray, on which at least two independent constraints are active. Such a ray is perpendicular to two independent original vectors, and hence is their cross product up to sign. This proves the exact finite test. □

Next write

$$D_W(z) = \det(W \operatorname{diag}(z) V) = \sum_{|I|=n-3} \det W_{:,I} \det V_{I,:} \prod_{i \in I} z_i. \tag{9}$$

If all coefficients vanish, every equilibrium is degenerate by (6) and (7). If the nonzero coefficients have one sign, there is at most one positive equilibrium per class. Indeed, if x, y are two such equilibria, $\log x - \log y = Vt$ and $x - y = \operatorname{diag}(\ell) Vt$ with all logarithmic means $\ell_i > 0$. Then $W \operatorname{diag}(\ell) Vt = 0$, contradicting (9) unless $t = 0$.

A further exact exclusion concerns the aggregated kinetic rank. Let K be a rational basis matrix of $\ker G$, write a general flux as $K\theta$, and sum its reaction vectors separately for each source, giving columns $L_1(\theta), \ldots, L_4(\theta)$ with sum zero. Project onto three independent coordinates. If the cubic polynomial

$$\det[L_1(\theta)\ L_2(\theta)\ L_3(\theta)]$$

vanishes identically, $\operatorname{rank} N < 3$ at every positive equilibrium of every subnetwork with this span. Such equilibria are degenerate. The certificate checks polynomial identity exactly. Nonzero evaluations are used only to retain a case, never to discard one.

The complete counts are:

	S1	S2	S3	S7	S9
All rank-three spans	4333	4287	4333	4397	23035
Missing a source	2965	3161	2965	3050	17734
No positive maximal flux	1126	782	1126	1042	4442
Injective	169	301	129	231	569
Class-degenerate	34	6	6	32	144
Kinetically degenerate	39	37	43	20	92
Remaining	0	0	64	22	54

Each span occurs in exactly one row after the first row. After fixing the ACR coordinate, remove its column from W and row-reduce. Permutations $A \leftrightarrow B$, $C \leftrightarrow D$, and interchange of the two pairs are allowed for S3 and S9; only $A \leftrightarrow B$ is needed for S7. The remaining spaces reduce to eleven, five and three forms respectively. The next sections list *all* of them and prove the required root and determinant properties.

5 The eleven S3 forms

The positive equilibria have

$$e = e_0, \qquad ab = p, \qquad cd = q, \qquad p, q, e_0 > 0.$$

After e is fixed the two conservation equations have one of the following forms. Constants T, U are arbitrary real numbers; existence of a positive solution supplies any needed positivity restrictions.

Index	First equation	Second equation
0	$b = T$	$c + d = U$
1	$a - 2c + 2d = T$	$b + c = U$
2	$a - c + d = T$	$b + (c + d)/2 = U$
3	$a - c + d = T$	$b + c = U$
4	$a - c + d = T$	$b + c + d = U$
5	$a - c + d = T$	$b + 2c = U$
6	$a - c/2 + d = T$	$b + c/2 = U$
7	$a = T$	$b + c + d = U$
8	$a + d = T$	$b + c = U$
9	$a + (c + d)/2 = T$	$b + (c + d)/2 = U$
10	$a + c + d = T$	$b + c + d = U$

Forms 0 and 7 fix $a, b > 0$ and $c+d$, so $cd = q$ is a quadratic on the interval $c, d > 0$. For form 8 take $a = t$, $b = p/t$, $c = U - p/t$, $d = T - t$. The positive interval is $p/U < t < T$, with $T, U > 0$. Multiplication of $cd - q = 0$ by $t > 0$ gives

$$-Ut^2 + (UT + p - q)t - pT = 0.$$

For forms 9 and 10, subtraction fixes $a - b$; together with $ab = p$ this uniquely fixes the positive pair (a, b), since $a - p/a$ is strictly increasing from $-\infty$ to $+\infty$. Again $c + d$ is fixed and $cd = q$ is quadratic on its positive interval.

For forms 1–6 put $c = t > 0$, $d = q/t$. Then

$$a = T + \alpha t - \beta q/t, \qquad b = U - \gamma t - \delta q/t,$$

where $\alpha, \beta, \gamma > 0$ and $\delta \geq 0$, as read directly from the table. The functions a, b are concave; a is strictly concave. Their common positivity set is an interval. On it,

$$(\log(ab))'' = \frac{a''}{a} - \frac{(a')^2}{a^2} + \frac{b''}{b} - \frac{(b')^2}{b^2} < 0.$$

Hence $ab = p$ has at most two roots, with opposite derivative signs when two roots exist.

For completeness, take the augmented determinant with rows R, $\nabla(ab - p)$, $\nabla(cd - q)$, where R is the conservation matrix in the table. On the displayed scalar curves its

multiplier of the derivative is, up to a fixed sign: $T > 0$ for forms 0 and 7; $t > 0$ for forms 1–6 and 8; and $h(a + b) > 0$ for forms 9 and 10, where $h = 1/2$ and $h = 1$ respectively. For form 8 multiplication of the scalar equation by t preserves the sign at a root. These identities also prove that repeated roots and identically zero reductions cannot be nondegenerate equilibria.

6 The five S7 forms

Here the positive equilibrium equations are

$$d = d_0, \qquad ab = pe, \qquad c^2 = qe, \qquad p, q, d_0 > 0.$$

The reduced variables are (a, b, c, e), and the conservation equations are:

Index	First equation	Second equation
0	$b + e = T$	$c - e = U$
1	$a - c + 2e = T$	$b + c = U$
2	$a - c/2 + e/2 = T$	$b + c/2 + e/2 = U$
3	$a - e = T$	$b = U$
4	$a + b + e = T$	$c = U$

For form 0 take $c = t$, $e = t - U$, $b = T - t + U$, $a = pe/b$. Positivity is an interval, and $c^2 - qe = t^2 - q(t - U) = 0$ is quadratic. Its determinant multiplier has magnitude $b > 0$.

For forms 1 and 2 put $u = 1/c > 0$, so $e = 1/(qu^2)$. Then

$$A(u) := a/c = Tu + \alpha - \frac{\beta}{qu}, \qquad B(u) := b/c = Uu - \gamma - \frac{\delta}{qu},$$

with $(\alpha, \beta, \gamma, \delta) = (1, 2, 1, 0)$ or $(1/2, 1/2, 1/2, 1/2)$. The positive interval is $A(u) > 0$, $B(u) > 0$. Both functions are concave and $A''(u) < 0$. Thus $\log(AB)$ is strictly concave, and $AB = p/q$ has at most two roots with opposite derivative signs. If $H = AB - p/q$, the augmented determinant with rows R, $\nabla(ab - pe)$, $\nabla(c^2 - qe)$ equals, up to fixed orientation,

$$q\big(H'(u) - 2H(u)/u\big).$$

At a root its multiplier of H' is the positive constant q.

For form 3 the equation $ab = pe$ is $UT + (U - p)e = 0$, with $U = b > 0$. There is at most one positive solution, unless the equation is an identity, in which case the positive solutions form a curve and are degenerate. For form 4, $c = U > 0$ fixes $e = U^2/q$; the sum and product of a, b are fixed, giving a quadratic on $a, b > 0$. The determinant multiplier has magnitude $q > 0$ in both scalar parametrizations used by the certificate.

7 The three S9 forms

The equilibrium equations are

$$e = e_0, \qquad ab = pf, \qquad cd = qf, \qquad p, q, e_0 > 0.$$

In variables (a, b, c, d, f) the three conservation equations have one of these forms:

Index	First equation	Second equation	Third equation
0	$b + f = T$	$c - f = U$	$d = V$
1	$a - c/2 + f/2 = T$	$b + c/2 + f/2 = U$	$d = V$
2	$a = T$	$b = U$	$c + d + f = V$

For form 0, put $f = t$, $b = T - t$, $c = t + U$, $d = V$, $a = pt/(T - t)$. Positivity is an interval with $V > 0$. The remaining equation is $(V - q)t + VU = 0$. It has at most one solution, or is an identity giving only degenerate equilibria. The determinant multiplier has magnitude $b > 0$.

For form 1, put $f = t$, $d = V > 0$, $c = qt/V$,

$$a = T + \tfrac{1}{2}(q/V - 1)t, \qquad b = U - \tfrac{1}{2}(q/V + 1)t.$$

The positive domain is an interval defined by three linear inequalities $t, a, b > 0$. The remaining equation $ab - pt = 0$ is quadratic. Its determinant multiplier has magnitude $V > 0$.

For form 2, $a = T > 0$ and $b = U > 0$ fix $f = TU/p$. The sum $c + d = V - TU/p$ and product $cd = qTU/p$ are fixed, so there are at most two positive solutions, with opposite derivative signs. The determinant multiplier has magnitude $p > 0$.

All nineteen forms have now been treated. Together with the source classification, the injectivity and degeneracy exclusions, and (6), this proves Theorem 2.

8 Reproduction, checks and limitations

From this directory, run:

```
python scripts/certify_sources.py
python scripts/certify_subspaces.py
python scripts/certify_normal_forms.py
python scripts/certify_five_example.py
```

The first script re-enumerates every eligible source set. The second generates from scratch all 40,385 reaction spans for the five specified source types, proves each exclusion, and writes the retained spaces and their classifications to `certificates/subspaces.json`. It does not read the old numerical candidate lists. The third symbolically checks all nineteen determinant identities and the concavity derivatives, including the conservation equations along every scalar curve. The fourth derives the upper-bound example from its reaction list, factors its complete steady-state equation and checks every stability claim in rational arithmetic. Each script prints PASS/FAIL and exits nonzero on a failure.

The environment used was Python 3.12, SymPy 1.14 and NumPy under Fedora 44 in WSL. The runs take less than one minute in total. Integer arrays use signed 64-bit entries; their input values are in $[-2, 2]$, three-minors have absolute value at most 48, adjugate entries at most 8, membership products at most 288, and cross-product inner products at most 48. Thus these operations cannot overflow. Nullspaces, polynomial identities and all subsequent coefficients are rational and symbolic. No numerical eigenvalue, optimizer tolerance, random search or sampling result is used as an exclusion.

A fresh-context verifier reimplemented the enumeration, obtained the same label for every span, and recorded two facts that make the proof sturdier; see `verification.md`. The positive-flux exclusion is not needed: every span it rejects is also rejected by the injectivity or degeneracy tests. The kinetic exclusion also has an elementary reason, without the cubic: on each span that it removes, the spans of the reaction vectors of some three sources fail Rado's condition, so three columns of N are forced to be dependent.

The primary source was read in its published HTML version. A public-source novelty audit on 31 October 2026, recorded in `references.md`, found no prior resolution of this exact-minimum question. Absence of search hits is not a proof of priority. The manuscript is a computer-assisted proof, checked by a fresh-context verifier, not by human referees. The five-reactant case is settled by (4). The result does not concern larger products or nonhyperbolic notions of stability. It establishes the exact reactant-complex minimum; it does not classify every network attaining it.

References

[1] Nidhi Kaihnsa, Tung Nguyen and Anne Shiu. Absolute concentration robustness and multistationarity in reaction networks: conditions for coexistence. *European Journal of Applied Mathematics*, **2024**.
DOI: 10.1017/S0956792523000335 ARXIV: 2307.04186 1, 2

[2] Claude Opus 5.5. Absolute concentration robustness with multistability in bimolecular networks: three species suffice, dimension two does not. chemistry-open-problems, result 27, 27 October. Local manuscript ../paper.tex; verified by a separate AI agent, not human-refereed. Includes the four-reactant source classification and the six-reactant example N1. **2026**. 1, 2, 5